



# Request Methods

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## HTTP?

Hyper Text Transfer Protocol (HTTP) is designed to enable communications between clients

and servers. It is a request-response protocol between a client and server.

Example: A client (browser) sends an HTTP request to the server; then the server returns a response to the client. The response contains status information about the request and may also contain the requested content.

## HTTP Methods

- GET
- POST
- PUT
- HEAD
- DELETE
- PATCH
- OPTIONS



# The GET Method

GET is used to request data from a specified resource.

Note that the query string (name/value pairs) is sent in the URL of a GET request:

```
/test/demo_form.php?name1=value1&name2=value2
```

## Some notes on GET requests:

- GET requests can be cached
- GET requests remain in the browser history
- GET requests can be bookmarked
- GET requests should never be used when dealing with sensitive data
- GET requests have length restrictions
- GET requests are only used to request data (not modify)

# The POST Method

POST is used to send data to a server to create/update a resource.

The data sent to the server with POST is stored in the request body of the HTTP request:

```
POST /test/demo_form.php HTTP/1.1
Host: w3schools.com

name1=value1&name2=value2
```

## Some notes on POST requests:

- POST requests are never cached
- POST requests do not remain in the browser history
- POST requests cannot be bookmarked
- POST requests have no restrictions on data length



|                             | GET                                                                                                                                                         | POST                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| BACK button/Reload          | Harmless                                                                                                                                                    | Data will be re-submitted (the browser should alert the user that the data are about to be re-submitted)       |
| Bookmarked                  | Can be bookmarked                                                                                                                                           | Cannot be bookmarked                                                                                           |
| Cached                      | Can be cached                                                                                                                                               | Not cached                                                                                                     |
| Encoding type               | application/x-www-form-urlencoded                                                                                                                           | application/x-www-form-urlencoded or multipart/form-data. Use multipart encoding for binary data               |
| History                     | Parameters remain in browser history                                                                                                                        | Parameters are not saved in browser history                                                                    |
| Restrictions on data length | Yes, when sending data, the GET method adds the data to the URL; and the length of a URL is limited (maximum URL length is 2048 characters)                 | No restrictions                                                                                                |
| Restrictions on data type   | Only ASCII characters allowed                                                                                                                               | No restrictions. Binary data is also allowed                                                                   |
| Security                    | <p>GET is less secure compared to POST because data sent is part of the URL</p> <p>Never use GET when sending passwords or other sensitive information!</p> | POST is a little safer than GET because the parameters are not stored in browser history or in web server logs |
| Visibility                  | Data is visible to everyone in the URL                                                                                                                      | Data is not displayed in the URL                                                                               |



## The PUT Method

PUT is used to send data to a server to create/update a resource.

The difference between POST and PUT is that PUT requests are idempotent. That is, calling the same PUT request multiple times will always produce the same result. In contrast, calling a POST request repeatedly have side effects of creating the same resource multiple times.

## The HEAD Method

HEAD is almost identical to GET, but without the response body.

In other words, if GET /users returns a list of users, then HEAD /users will make the same request but will not return the list of users.

A HEAD request is useful for checking what a GET request will return before actually making a GET request - a HEAD request can read the Content-Length header to check the size of the file, without actually downloading the file.

## The DELETE Method

The DELETE method deletes the specified resource.

## The PATCH Method

The PATCH method is used to apply partial modifications to a resource.

## The OPTIONS Method



## THE CONNECT METHOD

The CONNECT method is used to start a two-way communications (a tunnel) with the requested resource.

## The TRACE Method

The TRACE method is used to perform a message loop-back test that tests the path for the target resource (useful for debugging purposes).

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